USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Source of Inter-Domain TMU Parameters
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:
An accurate description of the source of the Inter-Domain TMU Parameters for Host Routers and Device Routers

Benefits as a result of the changes:
A consistent description that matches both Inter-Domain Link on a Host Router and a Device Router

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
None

Actual Change

(a). 7.3.2 – Inter-Domain Time Sync

To Text:

~~After calculating the Inter-Domain time offset, Inter-Domain frequency offset and Inter-Domain time stamp,~~ If the the Router with the IDTI Port is on a Host Router, it shall update the *InterDomainTimeStamp*, *FreqOffsetFromInterDomainHR* and *TimeOffsetFromInterDomainHR* fields in Router Configuration Space. If the IDTI Port is on a Device Router, it shall prepare an Inter-Domain Time Stamp Packet as described in Section 7.3.3.3 where:

(b). Table 7-8 – Follow-Up Packet Payload

To Text:

DW	Bits	Field	Description
	15:0	*IDTimeStamp*	For a Host Router: If the *IDE* bit is set to 1b, this field contains the most recent value of the Inter-Domain time stamp received from the IDTI Port. If the *IDE* bit is set to 0b, then this field shall be set to 0. For a Device Router: This field shall contain the *IDTimeStamp* value from the last Follow-Up Packet that the ~~IDTI Port~~Router received.
9	31:0		
10	31:0		
11	31:0	*TimeOffsetFromInterDomainHR*	For a Host Router: If the *IDE* bit is set to 1b, this field contains value of the *TimeOffsetFromInterDomainHR* field in the last received Inter-Domain Time Stamp Packet. If the *IDE* bit is set to 0b, then this field shall be set to 0. For a Device Router: This field shall contain the *TimeOffsetFromInterDomainHR* value in the last Follow-Up Packet that the ~~IDTI Port~~Router received.
12	31:0		
13	31:0	*FreqOffsetFromInterDomainHR*	For a Host Router: If the *IDE* bit is set to 1b, this field contains value of the *FreqOffsetFromInterDomainHR* field in the last received Inter-Domain Time Stamp Packet. If the *IDE* bit is set to 0b, then this field shall be set to 0. For a Device Router: This field shall contain the *FreqOffsetFromInterDomainHR* value in the last Follow-Up Packet that the ~~IDTI Port~~Router received.

(c). Table 8-4 – TMU Router Configuration Capability Fields

To Text:

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
16	TMU_RTR_CS_16	31:0	**Inter-Domain Time Stamp Low** For a Host Router~~n Inter-Domain Time Initiator~~: This field shall contain the least significant 32 bits of the ~~computed~~ value of the Inter-Domain time stamp received in the last Inter-Domain Time Stamp Packet.~~. The time stamp shall be computed as described in Section 7.4.2.1.~~ For ~~all other~~Device Router~~s~~: This field shall contain the least significant 32 bits of the IDTimeStamp value contained in the last received Follow-Up Packet.	RO	0
17	TMU_RTR_CS_17	31:0	**Inter-Domain Time Stamp Middle** For a Host Router~~n Inter-Domain Time Initiator~~: This field shall contain the middle 32 bits of the ~~computed~~ value of the Inter-Domain time stamp received in the last Inter-Domain Time Stamp Packet. ~~The time stamp shall be computed as described in Section 7.4.2.1.~~ For ~~all other~~Device Router~~s~~: This field shall contain the middle 32 bits of the IDTimeStamp value contained in the last received Follow-Up Packet.	RO	0
18	TMU_RTR_CS_18	15:0	**Inter-Domain Time Stamp High** For an Inter-Domain Time Initiator: This field shall contain the most significant 32 bits of the ~~computed~~ value of the Inter-Domain time stamp received in the last Inter-Domain Time Stamp Packet. ~~The time stamp shall be computed as described in Section 7.4.2.1.~~ For ~~all other~~ Device Router~~s~~: This field shall contain the most significant 32 bits of the IDTimeStamp value contained in the last received Follow-Up Packet.	RO	0